

Evaluating the Effectiveness of a Behaviour-Aware Multilingual Voice-Based Local Service Discovery System Using a Multi-Agent AI Framework

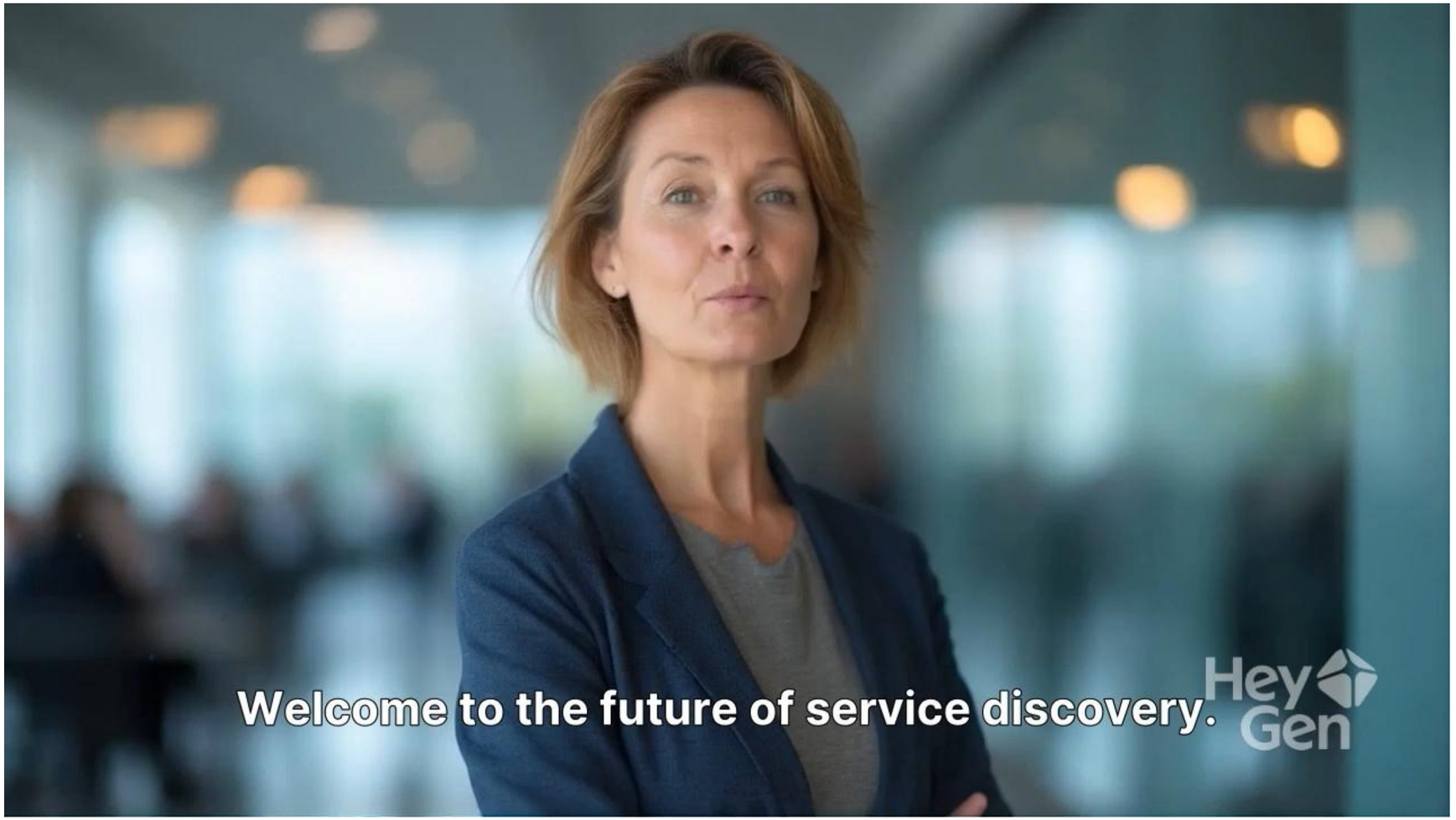


RESEARCH GROUP ID : R26-IT-109

PROGRESS PRESENTATION 2

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Video



Smart Accessible Learnina Environment



Problem

- ❖Mixed-Language Understanding: Difficulty understanding Sinhala-English and Singlish speech.
- ❖Incomplete Speech: Difficulty interpreting fragmented requests during urgent situations.
- ❖Limited Service Matching: Failure to consider urgency, location, distance, availability, and provider reliability together.
- ❖Unequal Job Distribution: Highly rated providers may receive more opportunities than newer or lower-skilled workers.

Motivation

- ❖Improve Accessibility: Enable natural communication through Sri Lankan mixed-language speech.Support
- ❖Urgent Requests: Detect intent and urgency from incomplete or fragmented speech.
- ❖Enable Smarter Matching: Use real-time context and multiple factors to select suitable providers.
- ❖Promote Fair Employment: Improve job distribution and create opportunities for newer and lower-skilled workers.

Adaptive Multilingual and Urgency-Aware Voice Intent Model



Proposed Solution

A behavior-aware, multilingual voice-based local service discovery system that understands Sinhala-English and Singlish mixed speech, including incomplete requests, while detecting the urgency of service requests. The system uses multiple AI agents to Analyse factors such as location, distance, availability, reliability, and urgency to identify and coordinate suitable service providers. It also provides workforce intelligence through demand prediction, performance analysis, and fairness-aware task allocation to support balanced job distribution and sustainable employment opportunities for service providers.

Key Features

- ❖ Multilingual Voice Interaction
- ❖ Incomplete Speech Understanding
- ❖ Urgency Detection
- ❖ AI-Based Service Matching
- ❖ Real-Time Provider Coordination
- ❖ Workforce Analytics
- ❖ Fair Job Allocation
- ❖ Demand Prediction

NON-Functional Goals

- ❖ High Performance
- ❖ Security & Privacy
- ❖ Accuracy
- ❖ Reliability
- ❖ Scalability
- ❖ Availability
- ❖ Fairness
- ❖ Maintainability

Adaptive Multilingual and Urgency-Aware Voice Intent Model



- Multilingual Speech Understanding – Sinhala-English and Tamil-English code-switching.
- Fragmented Speech Handling – Understands incomplete emergency requests.
- Intent Recognition – Identifies the service required from voice input.
- Urgency Detection – Classifies requests as Normal, Moderate, or High urgency using acoustic features.
- Image-Based Service Classification – Analyses uploaded images to identify the type/complexity of the service required.

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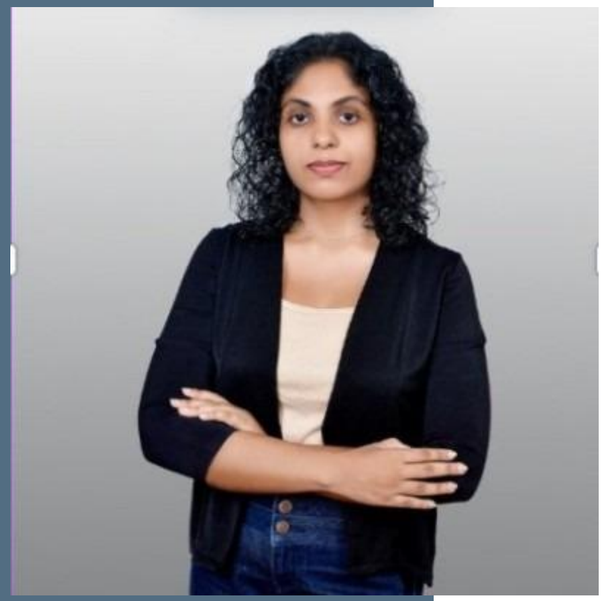
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Adaptive Multi-Agent Coordination Framework



- Context Agent – Analyses location, request time, and urgency.
- Search Agent – Finds suitable providers based on service category and location.
- Distance Agent – Calculates travel distance and estimated arrival time.
- Availability Agent – Checks whether providers are available and able to accept the job.
- Ranking Agent – Ranks providers using reliability, ratings, response speed, and urgent-task performance.

AI-Driven Employment Empowerment & Workforce Intelligence Framework



- Workforce Analytics Dashboard – Identifies underutilized workers, skill shortages, demand areas, and workforce performance trends.
- Fair Task Allocation – Reduces over-reliance on highly rated providers and creates opportunities for newer and lower-skilled workers.
- Skill Assessment & Training – Identifies skill gaps and recommends targeted training resources.
- Dynamic Reputation Model – Continuously evaluates completion rate, punctuality, delays, customer feedback, and urgent-task performance.
- Demand Prediction – Predicts future service demand by location, service type, and time period.
- Disaster-Aware Pre-Mobilization – Anticipates demand surges from events such as flooding and monsoon conditions and prepares relevant workers in advance.
- Income Stability Prediction – Analyses worker earning patterns and identifies potential income risks.

Overall Research Gap



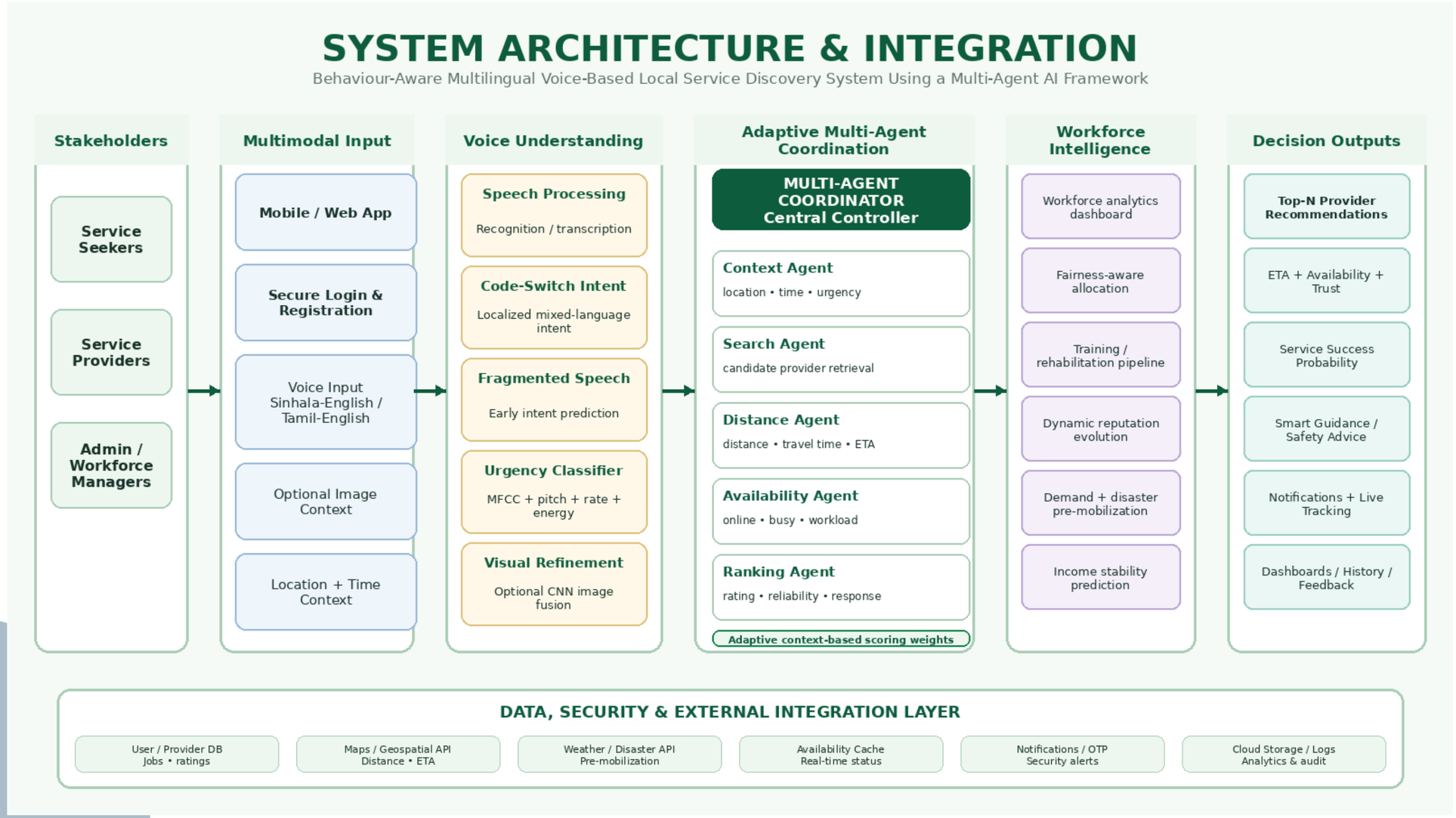
OVERALL RESEARCH GAP

Behaviour-Aware Multilingual Voice-Based Local Service Discovery System

Feature / Capability	Traditional Voice / Service Systems	Existing Local Service Apps	General Multi-Agent / Recommender Research	Proposed System
Mixed-language code-switch understanding (Sinhala-English / Tamil-English)	Limited	✗	Partial	✓
Fragmented / incomplete emergency speech handling	Limited	✗	Partial	✓
Acoustic urgency detection (MFCC, pitch, speech rate, energy)	✗	✗	Limited	✓
Context- and urgency-aware provider matching	Limited	Partial	Partial	✓
Specialized multi-agent provider coordination	✗	✗	Partial	✓
Real-time ETA + availability evaluation	Limited	Partial	Partial	✓
Dynamic context-based adaptive scoring weights	✗	✗	Limited	✓
Real-time monitoring and intelligent reassignment	✗	Partial	Limited	✓
Workforce fairness, demand & income intelligence	✗	✗	Limited	✓
One integrated intelligent service platform	✗	Partial	✗	✓

✓ Full support ✗ Not addressed Limited / Partial = incomplete or non-adaptive support

System Architecture & Integration



Commercialization



- **Target Users / Market** – Customers, local service providers, gig workers, service-platform administrators, and workforce managers in Sri Lanka.

- **Value Proposition** - AI-powered multilingual voice-based service discovery with urgency-aware matching, intelligent provider selection, fair workforce allocation, reputation management, and demand prediction.

- **Cost & Feasibility** – Developed as a web- and mobile-based system using open-source AI and software technologies, reducing development and licensing costs. The system can be deployed using cloud infrastructure and scaled according to user demand.

- **Adoption Plan** – Pilot testing with selected customers and service providers, followed by wider deployment through a web and mobile application and administrative platform across Sri Lanka.





THANK YOU !